



Sydney Boys High School

MOORE PARK
SURRY HILLS

DECEMBER 2003

HSC Assessment Task #1

YEAR 11

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time 90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for careless or badly arranged work.
- Start each question in a separate answer booklet.

Total Marks - 80 Marks

- Attempt Questions 1 to 5
- All questions are of equal value.

Examiner: *R. Boros*

Question 1: (16 marks)

Marks

- (a) Evaluate $\log_p 18$ given that $\log_p 3 = 0.4771$ and $\log_p 2 = 0.3010$. 2
- (b) Write a single logarithm for $\log x - \log y + 2 \log z$. 1
- (c) For what value of n is the sum of n terms of $12 + 15 + 18 + \dots$ equal to 441? 2
- (d) Evaluate $\sum_{n=3}^{13} 2^n$ 2
- (e) One card is drawn out from a set of cards numbered 1 to 20. Find the probability of drawing out an even number or a number less than 8. 2
- (f) When 2 regular dice are thrown and the total on these dice are counted, find the probability of scoring a total greater than 7. 2
- (g) A plant has a probability 0.7 of producing a variegated leaf. If 3 plants are grown, find the probability of producing no plants with variegated leaves. 3
- (h) A coin is tossed n times. Find an expression for the probability of throwing at least 1 tail. 2

Question 2: (16marks) START A NEW BOOKLET

Marks

(a) Simplify $\frac{(x^{m+1})^2 \times (x^3)^{n+1}}{x^{5m}}$. 2

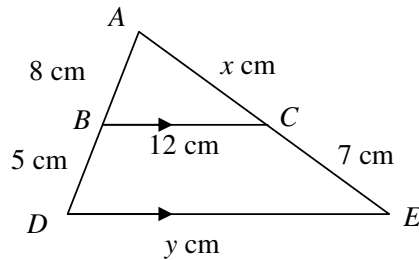
(b) Solve for x : $2^{x-1} = \frac{\sqrt{2}}{32}$ 2

(c) Write in simplest form: $\frac{2^{n+2} + 8}{2^{2n} + 2^{n+1}}$ 2

(d) Show that the points $A(6a, -2b)$, $B(2a, 0)$ and $C(0, b)$ are collinear. 2

(e) Prove that the points $A(3, 5)$, $B(4, 4)$, $C(1, 1)$ and $D(0, 2)$ are the vertices of a rectangle. 4

(f) Prove that $\triangle ABC \parallel \triangle ADE$. Hence find the values of x and y .



4

Question 3: (16 marks) START A NEW BOOKLET

Marks

- (a) Find $\lim_{x \rightarrow 1} \frac{x^2 + 2x - 3}{x - 1}$ 2
- (b) (i) Find the gradient of the tangent to the curve $y = x^2 + 2x + 1$ at the point (x, y) .
 (ii) Hence find the gradient of the tangent at the point $(\frac{1}{2}, 2\frac{1}{4})$. 3
 (iii) Find the angle which the tangent in (ii) makes with the positive direction of the x axis.
- (c) Find the first derivative of:
 (i) $y = \frac{-7}{x+1}$
 (ii) $y = (x^2 + x)^3$ 5
 (iii) $y = \frac{1}{\sqrt{3x^2 + 4}}$
- (d) Find the gradient of the normal to the curve $y = 5x\sqrt{4-x}$ at the point $(3, 15)$ 3
- (e) Find the maximum value of the function $y = x^2 - 4x + 3$ in the domain $1 \leq x \leq 4$. 3

Question 4: (16 marks) START A NEW BOOKLET

Marks

- (a) For the curve $y = 2x^3 - 3x^2 - 12x + 2$:
- (i) Find all stationary points.
 - (ii) Determine the nature of the stationary points. 9
 - (iii) Find any points of inflexion.
 - (iv) Sketch the curve.
- (b) Show that $y = \frac{5}{x}$ is always a decreasing function for all real $x \neq 0$. 2
- (c) Draw a neat sketch of a continuous curve $y = f(x)$ which has the following features:
- $f'(x) < 0$ for $0 \leq x < 3$
 - $f'(3) = 0$
 - $f'(x) > 0$ for $3 < x < 7$ 3
 - $f'(7) = 0$ and
 - $f'(x) > 0$ for $7 < x \leq 10$.
- (d) For a certain curve $y'' = x^2(x-1)^3(x-3)$, for what values of x is the curve concave up? 2

Question 5: (16 marks) START A NEW BOOKLET

Marks

- (a) Solve for x (correct to 2 decimal places): $2^x = 3^{x-1}$. 2
- (b) If $x^2 + y^2 = 7xy$, show that $\log(x + y) = \log 3 + \frac{1}{2}\log x + \frac{1}{2}\log y$. 2
- (c) A ball is dropped from a height of 1 metre and bounces to $\frac{2}{3}$ of its height on each bounce. What is the total distance travelled by the ball? 3
- (d) A sum of \$3 000 is invested at the beginning of each year in a superannuation fund. At the end of 35 years, how much money is available if the money invested earns interest at the rate of 6% per annum (compounded annually). 4
- (e) A sum of \$75 000 is borrowed at an interest rate of 12% per annum, monthly reducible. If the money is repaid at regular monthly intervals over 10 years, find the amount of each payment. 5



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SAMPLE SOLUTIONS

16

Question ①

$$(a) \log_p 18 = \log_p 3^2 + \log_p 2$$

$$= 2 \log_p 3 + \log_p 2$$

$$= 1.2552. \quad 2$$

$$(1) (b) \log x - \log y + 2 \log z$$

$$= \log \left(\frac{x z^2}{y} \right).$$

$$(c) S_n = \frac{n}{2} (2a + (n-1)d)$$

$$\therefore 441 = \frac{n}{2} [24 + (n-1)3]$$

$$\therefore 441 = \frac{3n}{2} (n+7)$$

$$\Rightarrow n^2 + 7n - 294 = 0$$

$$\therefore (n+21)(n-14) = 0$$

$$\therefore n = 14 \quad 2$$

$$(d) 2^3 + 2^4 + \dots + 2^{13}$$

$$n=11, a=2^3, t=3.$$

$$\therefore S_{11} = 8(2^{11} - 1)$$

$$= 8 \times 2047$$

$$= 16376. \quad 2$$

$$(e) P(\text{even}) = \frac{1}{2}.$$

$$P(x < 8) = \frac{7}{20}.$$

$$\therefore P(E) \text{ or } P(x < 8)$$

$$= \frac{1}{2} + \frac{7}{20} - \frac{3}{20}$$

$$= \frac{14}{20}$$

$$= \frac{7}{10} \quad 2$$

$$(f) S = 6 \times 6 = 36$$

$$Sum > 7.$$

$$(2,6) (6,2) (3,5) (5,3)$$

$$(3,6) (6,3) (4,4) (4,5)$$

$$(5,4) (4,6) (6,4) (5,5)$$

$$(5,6) (6,5), (6,6).$$

$$\therefore P(\text{sum} > 7) = \frac{15}{36}$$

$$= \frac{5}{12}. \quad 2$$

$$(g) (0.3)^3 = 0.027.$$

(3).

(R)

$$P(\text{at least 1 tail})$$

$$= 1 - Pr(\text{no tail})$$

$$= 1 - \frac{1}{2^n}$$

2

$$\begin{aligned}
 \boxed{2} \quad 2. \quad (a) \quad \frac{x^{2m+2} \times x^{3n+3}}{x^{5m}} &= \frac{x^{2m+3n+5}}{x^{5m}}, \\
 &= x^{3n-3m+5}.
 \end{aligned}$$

Or (if you realised it was a typo)—

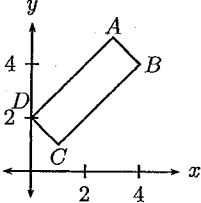
$$\begin{aligned}
 \frac{(x^{m+1})^2 \times (x^3)^{m+1}}{x^{5m}} &= \frac{x^{2m+2} \times x^{3m+3}}{x^{5m}}, \\
 &= x^5.
 \end{aligned}$$

$$\begin{aligned}
 \boxed{2} \quad (b) \quad 2^{x-1} &= 2^{1/2} \times 2^{-5}, \\
 x-1 &= -4\frac{1}{2}, \\
 x &= -3\frac{1}{2}.
 \end{aligned}$$

$$\begin{aligned}
 \boxed{2} \quad (c) \quad \frac{2^{n+2} + 2^3}{2^{2n} + 2^{n+1}} &= \frac{2^2(2^n + 2)}{2^n(2^n + 2)}, \\
 &= 2^{2-n}.
 \end{aligned}$$

$$\begin{aligned}
 \boxed{2} \quad (d) \quad \text{Slope } AB &= \frac{0 - -2b}{2a - 6a}, & \text{Slope } BC &= \frac{b - 0}{0 - 2a}, \\
 &= \frac{2b}{-4a}, & &= -\frac{b}{2a}, \\
 &= -\frac{b}{2a}. & &= \text{Slope } AB.
 \end{aligned}$$

As B is common, ABC is a straight line.

$\boxed{4} \quad (e)$


$$\begin{aligned}
 m_{AB} &= -1, & \text{so } m_{AB} \times m_{BC} &= -1, \\
 m_{BC} &= 1, & m_{BC} \times m_{CD} &= -1, \\
 m_{CD} &= -1, & m_{CD} \times m_{DA} &= -1, \\
 m_{DA} &= 1. & m_{DA} \times m_{AB} &= -1.
 \end{aligned}$$

$\therefore$ All vertices are right angles and $ABCD$ is a rectangle.

$$\begin{aligned}
 \boxed{4} \quad (f) \quad \hat{A} &\text{ is common,} \\
 \hat{ABC} &= \hat{ADE} \text{ (corresponding angles, } BC \parallel DE), \\
 \therefore \triangle ABC &\parallel \triangle ADE \text{ (equiangular).} \\
 \frac{x}{x+7} &= \frac{8}{13}, & \frac{y}{12} &= \frac{13}{8}, \\
 13x &= 8x + 56, & 2y &= 39, \\
 5x &= 56, & y &= 19\frac{1}{2}, \\
 x &= 11\frac{1}{5}
 \end{aligned}$$

Question 3

a) $\lim_{x \rightarrow 1} \frac{x^2 + 2x - 3}{x - 1}$

$= \lim_{x \rightarrow 1} \frac{(x-1)(x+3)}{x-1}$

$= \lim_{x \rightarrow 1} (x+3)$

$= 4$

[2]

b) $y = x^2 + 2x + 1$

(i) $y' = 2x + 2$

[1]

(ii) At $(\frac{1}{2}, 2\frac{1}{2})$ $y' = 2(\frac{1}{2}) + 2$
 $= 3$

$\therefore \text{Gradient} = 3$

[1]

(iii) $m = 3 = \tan \alpha$

Where α is the angle of inclination

$\therefore \alpha = \tan^{-1} 3$

$= 71^\circ 34'$

[1]

c) (i) $y = \frac{-7}{x+1}$

$= -7(x+1)^{-1}$

$y' = 7(x+1)^{-2} \times 1$

$= \frac{7}{(x+1)^2}$

[1]

(ii) $y = (x^2 + x)^3$

$y' = 3(x^2 + x)^2 \cdot \frac{d}{dx}(x^2 + x)$

$= 3(x^2 + x)^2 (2x + 1)$

$= (6x + 3)(x^2 + x)^2$

$= 3x^2(2x + 1)(x + 1)^2$

(iii) $y = \frac{1}{\sqrt{3x^2 + 4}}$

$y' = \frac{-1}{(\sqrt{3x^2 + 4})^2} \times \frac{d}{dx} \sqrt{3x^2 + 4}$

$= \frac{-1}{3x^2 + 4} \times \frac{1}{2\sqrt{3x^2 + 4}} \cdot \frac{d}{dx}(3x^2 + 4)$

$= \frac{-6x}{2(3x^2 + 4)\sqrt{3x^2 + 4}}$

$= \frac{-3x}{(3x^2 + 4)^{3/2}}$

[2]

(d) $y = 5x\sqrt{4-x}$

$y' = 5\sqrt{4-x} + 5x \cdot \frac{-1}{2\sqrt{4-x}}$

$= 5 \left[\sqrt{4-x} - \frac{x}{2\sqrt{4-x}} \right]$

At $x=3$ Grad. of $y' = 5 \left[1 - \frac{3}{2 \times 1} \right]$

$= 5 \left[-\frac{1}{2} \right]$

$= -\frac{5}{2}$

$\therefore \text{Grad. of Normal} = \frac{2}{5}$

[3]

(e) $y = x^2 - 4x + 3$

Function has a minimum.

$\therefore$ Max Value at boundary.

$y(1) = 1 - 4 + 3$

$= 0$

$y(4) = 16 - 16 + 3$

$= 3$

$\therefore \text{Max Value} = 3$

[3]

QUESTION 4

(a) $y = 2x^3 - 3x^2 - 12x + 2$

(i) Stat. pts where $y' = 0$

$\therefore y' = 6x^2 - 6x - 12$

i.e. $y' = 6(x-2)(x+1) = 0$ 3
when $x=2$ and $x=-1$

$\therefore (2, -18)$ and $(-1, 9)$

(ii) $y'' = 12x - 6$

At $x=2$, $y'' = 18 > 0$

$\therefore \text{MIN}(2, -18)$ 1

At $x=-1$, $y'' = -18 < 0$

$\therefore \text{MAX}(-1, 9)$ 1

(iii) For P.O.I. $f''(x) = 0$ and
there must be a change of concavity,
i.e. $f''(x)$ must change sign.

Now $f''(x) = 0$ when $x = \frac{1}{2}$ 1

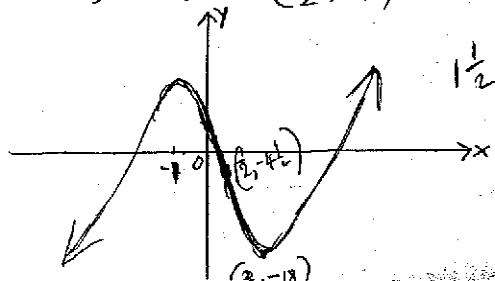
and

x	$< \frac{1}{2}$	$> \frac{1}{2}$
$f''(x)$	$-$	$+$

 1

$\therefore$ at $x = \frac{1}{2}$ curve changes,
concavity from c.d. to c.u.

(iv) $\Rightarrow$ P.O.I. at $(\frac{1}{2}, -4\frac{1}{2})$



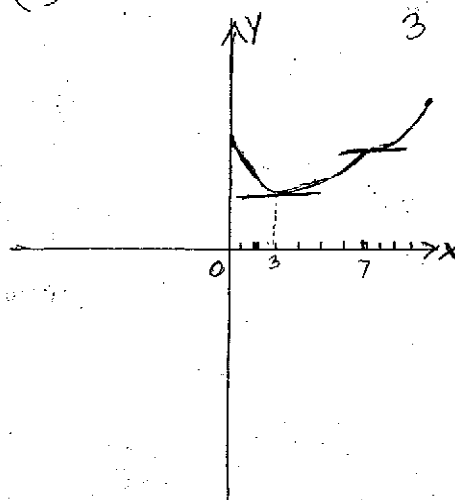
(b) $y = \frac{5}{x}$ 2

$\frac{dy}{dx} = -\frac{5}{x^2} < 0$

for all real x where $x \neq 0$

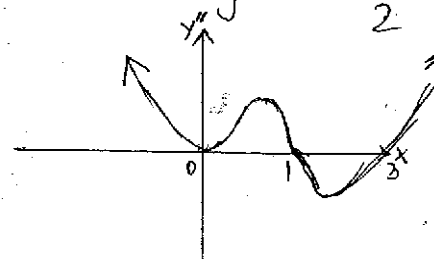
$\therefore y = \frac{5}{x}$ is decreasing for
these values of x .

(c)



(d) $y'' = x^2(x-1)^3(x-3)$

c.u. when $y'' > 0$ 2



$\therefore$ c.u. when

$x < 1$ and $x > 3$

Note: $x \neq 0$

Question 5

$$\begin{aligned}
 \text{(a)} \quad 2^x &= 3^{x-1} \\
 \log_{10} 2^x &= \log_{10} 3^{x-1} \\
 x \log_{10} 2 &= (x-1) \log_{10} 3 \\
 x \log_{10} 2 - x \log_{10} 3 &= -\log_{10} 3 \\
 x(\log_{10} 2 - \log_{10} 3) &= \log_{10} 3^{-1} = \log_{10} 1/3 \\
 x(\log_{10} 2/3) &= \log_{10} 1/3 \\
 x &= \frac{\log_{10} 1/3}{\log_{10} 2/3} \approx 2.71
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad x^2 + y^2 &= 7xy \quad x^2 + y^2 + 2xy = 9xy \\
 x^2 + y^2 + 2xy &= (x+y)^2 = 9xy \\
 \log(x+y)^2 &= \log 9xy \\
 2 \log(x+y) &= \log 9 + \log x + \log y = \log 3^2 + \log x + \log y \\
 2 \log(x+y) &= 2 \log 3 + \log x + \log y \\
 \log(x+y) &= \log 3 + \frac{1}{2} \log x + \frac{1}{2} \log y
 \end{aligned}$$

QED

$$\begin{aligned}
 \text{(c)} \quad \text{Distance} &= 1 + 2 \left(\frac{2}{3} - 1 \right) + 2 \left(\frac{2}{3} - \frac{2}{3} \right) + 2 \left(\frac{2}{3} - \left(\frac{2}{3} \right)^2 \right) + \dots \\
 &= 1 + 2 \left(\frac{2}{3} + \left(\frac{2}{3} \right)^2 + \left(\frac{2}{3} \right)^3 + \dots \right) \\
 &= 1 + 2 \frac{\frac{2}{3}}{1 - \frac{2}{3}} = 1 + 2 \frac{\frac{2}{3}}{\frac{1}{3}} = 1 + 2 \cdot 2 = 5
 \end{aligned}$$

(d) The first \$3000 would earn $3000(1.06)^{35}$, the next \$3000 would earn $3000(1.06)^{34}$ and so on until the start of the 35th year where the last \$3000 would earn $3000(1.06)$.

So the total investment is worth

$$\begin{aligned}
 S &= 3000(1.06)^{35} + 3000(1.06)^{34} + \dots + 3000(1.06) \\
 S &= 3000 \left((1.06) + (1.06)^2 + \dots + (1.06)^{35} \right) \\
 &= 3000 \frac{1.06 \left((1.06)^{35} - 1 \right)}{1.06 - 1} \\
 &= \$354362.60
 \end{aligned}$$

(e) Let \$M be the monthly repayment, let \$A_n be the amount owing after n months.

12% pa = 1% per month, 10 years = 120 months

$$A_1 = 75000(1.01) - M$$

$$A_2 = A_1(1.01) - M$$

$$= 75000(1.01)^2 - M(1 + 1.01)$$

$$A_3 = A_2(1.01) - M$$

$$= 75000(1.01)^3 - M(1 + 1.01 + 1.01^2)$$

$$A_n = 75000(1.01)^n - M(1 + 1.01 + \dots + 1.01^{n-1})$$

$$A_{120} = 75000(1.01)^{120} - M(1 + 1.01 + \dots + 1.01^{119})$$

$$\text{Let } S_{120} = 1 + 1.01 + \dots + 1.01^{119}$$

$$= \frac{1.01^{120} - 1}{1.01 - 1} = 100(1.01^{120} - 1)$$

$$A_{120} = 0$$

$$M = \frac{75000(1.01)^{120}}{S_{120}} = 1076.03$$

So the monthly repayment is \$1076.03